

The Compiler Closure of TFPT

What concretely simplifies, reduces and closes relative to the seven original papers — a status assessment and reading guide

June 11, 2026

In one sentence

The theory tips from “many sectors with surprising hits” (the state of Papers 1–7) to “*one small discrete compiler generates those sectors*”: E_8 — which in v4.5 appeared only *peripherally* (the orbit cascade for scale ordering) — is now *built* from two TFPT atoms and moved to the centre; the flavor matrix, the fine-structure constant and the scale grammar follow as *fixed points* from exactly two inputs $c_3 = \frac{1}{8\pi}$ and $g_{\text{car}} = 5$.

The TFPT document set — what is treated where

Plain language: TFPT is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard Model, the constants and the scale grammar. The development is **five short documents plus Appendix H**, best read in order (this is the entry point; a detailed map is in §1):

1. **introduction** — reading guide, compiler closure, paper-by-paper comparison, predictions, the dependency DAG and proof ledger.
2. **tfpt_1_architecture_e8** — the two axioms, the derivation map, the EM fixed point α^{-1} , the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction.
3. **tfpt_2_standard_model** — the SM in one φ_0 -ladder formula, flavor from parabolic transport, the worked closures, and gravity/QG as the seam response.
4. **tfpt_3_e8_audit_bootstrap** — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
5. **tfpt_4_frontier** — honest status of η_B , m_p/m_e , Koide, dark matter, full QG.
- H. **tfpt_horizon_readouts** — **Appendix H** (unit-system reframe, *not* new physics): one seam constant as the universal horizon thermal code.
- O. **origin_theory** — **Origin Theory**: why no free fundamental number remains. The $(g_{\text{car}}, N_{\text{fam}})=(5, 3)$ skeleton, the triply-forced 8 (geometry = lattice = gravity), the order-30 Coxeter cycle, one boundary transport for flavor and horizon, and the gapped *unique attractor*. [I]/[L] core (v53–v56) plus one honestly-typed [P] cyclic interpretation.

You are reading document #1.

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Reading the markers (refined). The former overloaded “[M]” is split into the grade it actually carries, so a reviewer can see at a glance which claims are exact, which are formal, and which are conditional: **[I]** *exact identity* (e.g. $240 = 16 \cdot 5 \cdot 3$); **[L]** *Lie/lattice theorem* (e.g. $D_5 \oplus A_3 + \mu_4 = E_8$); **[F]** *formalised* (Lean- or script-verified); **[N]** *numerical fixed point* (reproduced, stable); **[P]** *physical/conditional* (follows under named hypotheses); **[A]** *axiom or open*.

Physical — under hypotheses	RG masses, QG QG.AMB.01
Numerical — fit-free fixed point	$\alpha^{-1}, \Lambda, \theta_{12}$ EM.FP.01
Lattice — Lie / glue / root system	$E_8 = (D_5 \oplus A_3) + \mu_4$ E8.GLU.01
Formal — Lean / exact script	P2 algebra FORM.P2.02
Axiom — declared inputs	P1 c_3 , P2 g_{car} AX.P1.01

The *claim stack* (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is **verification/status_ledger.csv** (Fig. 1).

No-free-pattern rule. To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* **[I]** (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

Admissible-invariant classes (the harder filter). Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function $e_k(a)$ of the anchor $a = (1, 1, 2)$; (ii) a power sum $p_n(a)$ or difference $p_{n+1} - p_n$; (iii) a determinant, spectrum, Smith normal form or trace of (R, Q, K, L) ; (iv) an anchor quadratic form $u^\top M v$ with $u, v \in \{1, a\}$; (v) a Plücker coordinate of the anchor plane $\langle 1, a \rangle$; (vi) an E_8 Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* (“look, another number”) into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

Reviewer path (read in this order; the rest is support)

1. the architecture: the two axioms $\{c_3, g_{\text{car}}\}$ and the two-engine picture (§1, the figures);
2. the E_8 glue $E_8 = (D_5 \oplus A_3) + \mu_4$ — document 1, Part II, machine-checked

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(verification/v1_e8_glue.py);
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3. the α^{-1} fixed point plus ablation (v3_em_alpha.py);
4. the **single status ledger** verification/status_ledger.csv — the source of truth for every claim’s grade, location, dependency and script;
5. the honest frontier list (document 4).

Everything carries a claim ID (e.g. E8.GLU.01, EM.FP.01) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

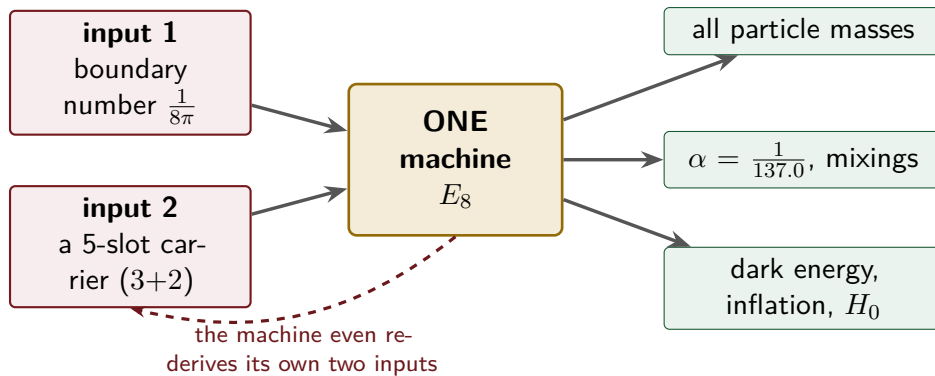
1 What this is about: the jump in construction principle

In plain words — what we found, and why it matters

What we found. The older version of the theory already reproduced dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs. What it could *not* say was *why* the same handful of small integers (2, 3, 5, 16, ...) kept reappearing in every sector. The new result closes exactly that gap: those numbers are not separate lucky hits. A single tiny “machine” — fed by just **two inputs** (a boundary number $\frac{1}{8\pi}$ and a five-slot carrier) — builds the exceptional symmetry E_8 , and E_8 is the unimodular “audit hull” that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not “printed” for free — where a scheme or analytic interface is needed, it is named).

How this drastically simplifies the theory. Instead of seven separate derivations that happen to share numbers, there is now *one generator and a handful of corollaries*. Roughly 250 pages of sector-by-sector work collapse to “two inputs + one machine”; the number of independent structural assumptions drops to **two**.

How plausible it is. A theory becomes more believable when it *explains more from less*. Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed loop below), so there is *no free dial left to tune*. The only genuinely free, continuous ingredient that remains is the number π .

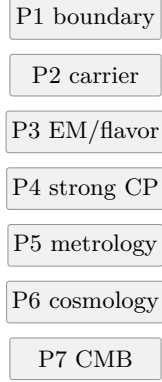


Two numbers go in on the left; the discrete Standard-Model core, the dimensionless constants and several scale readouts come out on the right — and the dashed loop is the point: the machine reproduces the very two numbers it started from, so the discrete core is overdetermined rather than fitted. (Dimensionful EW/QCD masses, the full quantum-gravity measure and the frontier items stay typed-open — see the status ledger.)

The seven original papers (tfpt-45, Papers 1–7) deliver a long chain of physical closures: boundary kernel → carrier → Standard-Model packet → EM fixed point → flavor transport →

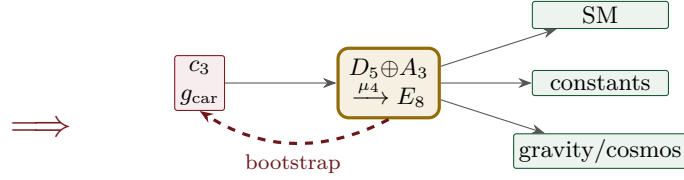
admissibility/strong CP $\rightarrow$ geometric gravity/metrology $\rightarrow$ cosmology/CMB. Each stage stands on its own and hits a remarkable number of values. What was missing was a *meta-statement* explaining *why* so many independent sectors use the same small stock of numbers (2, 3, 5, 16, 40, 41, 48, 240, 248).

Version 4.5 — seven papers



many parallel
sector closures;
 E_8 only a **periph-**
eral backdrop

New — one compiler



two inputs build E_8 ;
 E_8 feeds back and fixes
the inputs (closed loop)

The difference in one picture. v4.5 is a *stack of seven independent closures* that happen to share numbers, with E_8 assumed in the background. The new approach is *one discrete compiler*: two inputs build E_8 , E_8 outputs the sectors, *and* E_8 feeds back to fix the two inputs (the red loop). What were seven parallel results become one generator with corollaries; what was a posited E_8 becomes a constructed-and-self-checking closure.

Sharper still: one anchor $a = (1, 1, 2)$, and E_8 as a *compiler hull* [1]/

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity / splitting type $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$):

$$e_1(a) = 4 = |\mu_4|, \quad e_2(a) = 5 = g_{\text{car}}, \quad e_3(a) = 2 = |\mathbb{Z}_2|, \quad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums* $p_n(a) = 1^n + 1^n + 2^n = 2 + 2^n$ generate the big Lie data directly: $p_1=4=|\mu_4|$, $p_2=6=|R^+(A_3)|$, $p_3=10=\mathcal{A}_\Lambda$, and

$$|R(E_8)| = p_1 p_2 p_3 = 240, \quad \dim E_8 = p_1 p_2 p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \text{rank } E_8,$$

with the binary ladder $p_{n+1} - p_n = 2^n$. So the inputs collapse from $\{c_3, g_{\text{car}}\}$ to $a = (1, 1, 2)$ plus π . [verification/v23_anchor_generator.py]

And a is itself *half-forced* ([L]/[P]): on $\mathbb{P}^1$ every bundle splits (Birkhoff–Grothendieck), so once $|\mu_4| = 4$, $N_{\text{fam}} = 3$, $\deg E = -4$ are set, the three positive anti-degrees are positive integers summing to 4, and $4 = 2 + 1 + 1$ is the *unique* partition into three positive parts. The genuine remaining axiom is therefore not “why $a = (1, 1, 2)$ ” but “why the carrier $g_{\text{car}} = 5 / (|\mu_4|, N_{\text{fam}}) = (4, 3)$ ” — that interface (with c_3) is the [A] input layer.

What E_8 is (and is not). In TFPT E_8 is *not* an unbroken physical gauge group: it is the unimodular *audit* / *compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct E_8 gauge theory; the Lorentz signature appears only after projection; fermions are not “everything in the adjoint”. This is exactly why the known no-go results against literal E_8 world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain E_8 as a spacetime gauge symmetry, which

TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor $a = (1, 1, 2)$ into the E_8 audit hull.*

In one line:

The anchor generates the syntax; the boundary generates the scale; E_8 checks the consistency.

i.e. the discrete content (carrier $D_5 \oplus A_3 + \mu_4$, the operators R, Q, K, L) flows from $a = (1, 1, 2)$; the continuous content ($c_3 = \frac{1}{8\pi}$, φ_0 , α^{-1} , the exponential scales) flows from π ; and E_8 is the unimodular *checksum container*, not the world group.

That is now the case. The whole development is consolidated into **five companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the earlier notes of its layer into self-contained parts:

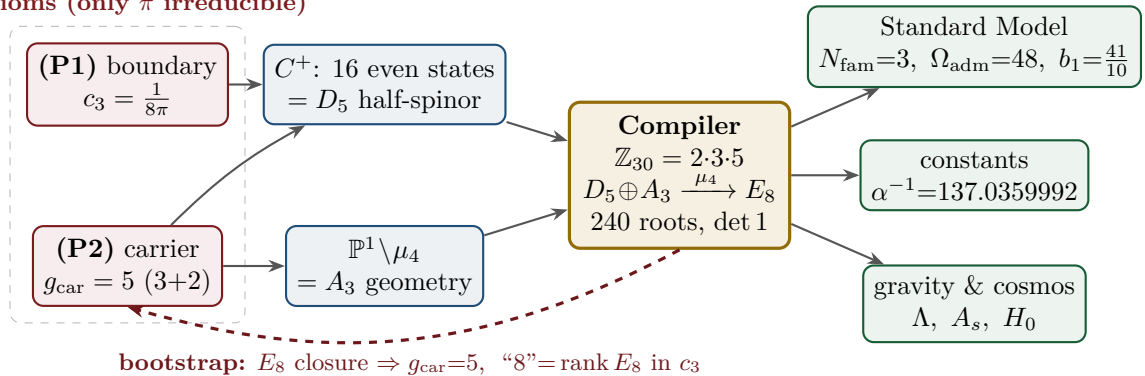
Document	What it contains (former notes now as parts)
tfpt_1_architecture_e8	Core. The two axioms $\{c_3, g_{\text{car}}\}$, the derivation map and the EM fixed point (Part I); the Coxeter-cyclotomic compiler $\mathbb{Z}_{30}=2\cdot3\cdot5$, the explicit $D_5 \oplus A_3 + \mu_4$ E_8 construction, the machine-checkable E_8 appendix and the group-level glue $(\text{Spin}(10) \times \text{SU}(4)) / \Delta \mathbb{Z}_4$ (Part II).
tfpt_2_standard_model	Standard Model. The full SM in one φ_0^{ret} -ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/ n_s, r , Starobinsky M , derived θ_{12} , quark c , the H2 splitting type, Λ -typing (Part III).
tfpt_3_e8_audit_bootstrap	E_8 audit & bootstrap. The seven E_8 slices as an audit raster ($E_6 \times A_2$ reads the flavor matrix) (Part I); the old cascade $D=60-2n$ as the same even-integer spine (Part II); the Möbius bootstrap — $g_{\text{car}}=5$ and the “8” in c_3 as overdetermined fixed points, only π irreducible (Part III).
tfpt_4_frontier	Frontier. The honest status of η_B , m_p/m_e , Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is <i>not</i> forced onto the ladder.
tfpt_horizon_readouts (App. H)	Appendix H — horizon unit system (reframe, not new physics). One seam constant $c_3 = \frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\text{GW}}=c$ — standard physics in seam units, two compiler fingerprints ($1920= W(D_5) $, $ \mu_4 =4$).

The five formerly “research-level” problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure $(\text{Spin}(10) \times \text{SU}(4)) / \Delta \mathbb{Z}_4$ in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the θ_{12} texture and the quark c in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a U_{point} anchor; θ_{12} the seam-misalignment residual). Five companion files (plus this introduction) instead of a dozen, each result where it logically lives.

2 The architecture at a glance

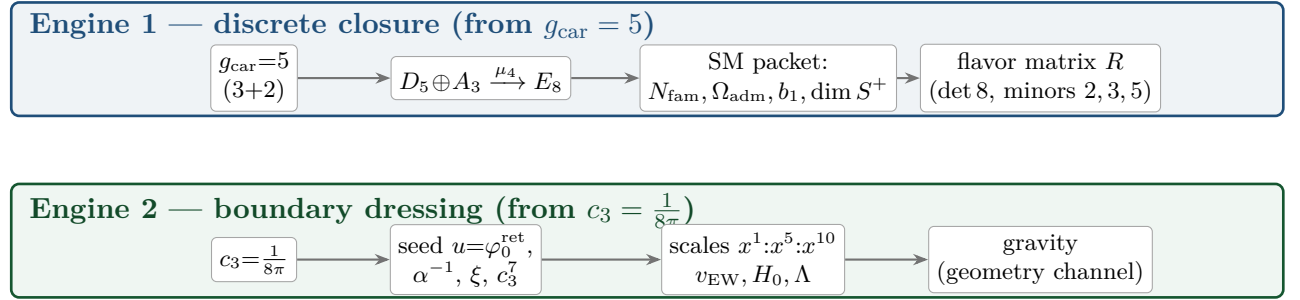
The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.

2 axioms (only π irreducible)



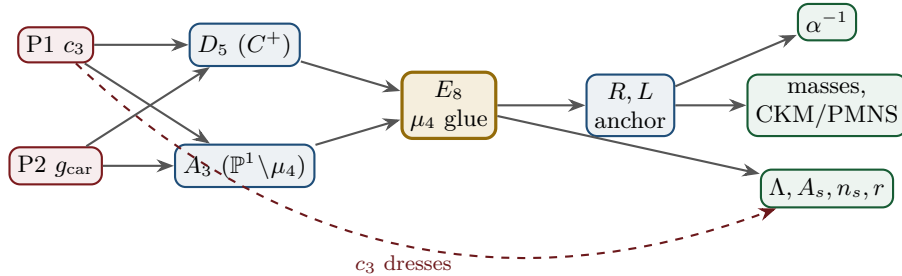
Key statement (the loop closes): previously D_5 , A_3 and E_8 sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the E_8 closure feeds back and *fixes the inputs*. $g_{\text{car}} = 5$ is the unique rank-fill/Coxeter/Pascal solution ($2^{g-1} = \binom{g}{0} + \binom{g}{1} + \binom{g}{2}$), and the integer 8 in $c_3 = \frac{1}{8\pi}$ is rank $E_8 = h(D_5) = \det R$. **Inputs and output are mutually locked — they prove each other;** the only thing *not* produced by the loop is the continuous π (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed self-consistency loop.

The master story is two engines. Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.



3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



Node	Definition	Inputs	St.	Output	Failure mode
P1	RP boundary kernel, $c_3=1/(8\pi)$	— (axiom)	[A]	c_3	no reflection-positive seam
P2	5-slot carrier $g_{\text{car}}=5$	— (axiom)	[A]	carrier 3+2	wrong family/charge lattice
D_5	C^+ even-Hamming = half-spinor 16	P2	[I]	$\dim S^+=16, Y$	group/matter mismatch
A_3	$\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.	P1	[I]	$N_{\text{fam}}=3, \mathbb{Z}_6$	wrong multiplicity
E_8	μ_4 glue: $\text{disc}=\mathbb{Z}_4, q\text{-sum}=2$	D_5, A_3	[I]	240, 248	not even-unimodular
R, L	residue + winding matrix, anchor (1, 1, 2)	A_3	[I]	$\det 8,$ minors (2, 3, 5)	wrong D_6 branch
α^{-1}	root of $F_{U(1)}(\alpha)=0$	$c_3, M=41$	[N]	137.0359992	no/second root (excl.)
masses	φ_0^{ret} -ladder $\lambda_Y^L \Lambda$	R, L, c_3	[I]/[P]	9 masses, mixings	hierarchy mismatch
θ_{12}	TBM + seam $\varepsilon=c_3$	R, L	[N]/[P]	0.307	seam-misalign. lemma fails
Λ, A_s	seam det. / R^2 scalaron	c_3	[P]	Λ, A_s, n_s, r	ambient RP fails

Marker key: **[I]** exact identity · **[L]** Lie/lattice theorem · **[F]** formalised · **[N]** numerical fixed point · **[P]** physical/conditional · **[A]** axiom / open.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

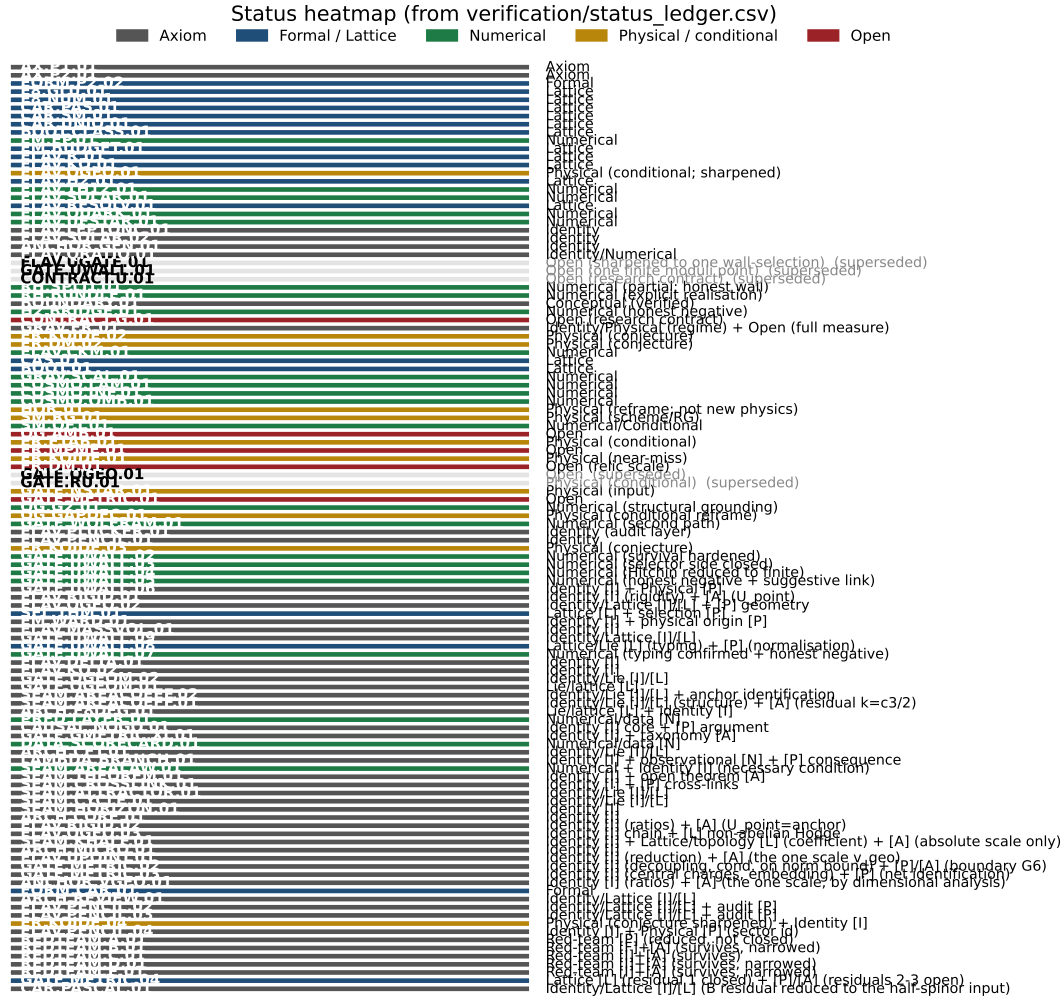


Figure 1: Status heatmap, generated directly from the single source of truth `verification/status_ledger.csv`. Grey = axiom, blue = formal/lattice, green = numerical, gold = physical/conditional, red = open. Each row carries a claim ID (`E8.GLU.01`, ...) that resolves to a ledger row with its dependencies and verification script. The ledger is *versioned*: each row has `active` and `canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early “flavor gate open” rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive and drawn dimmed — the canonical status is the only one read as current.

The compact status formula (three honest layers)

compiler	admissible IR physics	strict physical TOE
closed	conditionally closed (RP, gap)	open

TFPT 5.0 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the $R + R^2$ spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the R -bridge amplitude normalisation U_{point} *reduces to* v_{geo} (Gate 1 closed, v75 — the same anchor as $1/G$),

and the ambient metric measure is *holographically reduced* to a finite seam-boundary measure (Gate 2 tier B, v76) — which is in turn the rigorously-constructed $(E_8)_1$ lattice net (v77), so G_{metric} is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale v_{geo} — the *dimensional-analysis floor* that no theory escapes (v78), shared by flavor and gravity — the boundary net identification, and the named QFT/cosmology transfer interfaces (F_{frontier}).

Two points that are *not* in the residual (closed, but easy to mis-list as open).

(i) *Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ($\Delta = 6 \log \frac{3}{2} > 0$), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([I]/[L] for the attractor, [verification/v56_unique_attractor.py]; the *physical identification* of the transport operator stays [P]), reinforcing the static bootstrap web ($g_{\text{car}} = 5$ forced three ways *given* the E_8 closure rank — a consistency loop, see red team Target B — [verification/v6_bootstrap.py], [verification/v14_carrier_uniqueness.py]). (ii) *Newton’s constant is not an independent input:* fixing the physical Λ branch ($(8\pi)^2 \delta_{\text{top}} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\text{Pl}}$ *relative to* Λ , so G_N is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([I]/[P], [verification/v60_lambda_metrology_branch.py]), not predicted from pure numbers. What stays irreducible is then just π , the discrete E_8 -closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale v_{geo} — and that single scale is shared by *both* the flavor sector ($U_{\text{point}} \rightarrow v_{\text{geo}}$, v75) and gravity ($1/G$, v68): the two $[A]$ anchors are *one* (dimensional analysis; no metre from pure mathematics).

The hard reduction: everything open is two research gates plus typed interfaces

After the compiler closure, the entire residual is

$$\text{Rest} = (U_{\text{wall}}) \oplus (G_{\text{metric}}) \oplus (F_{\text{frontier}})$$

— one parabolic wall-selection, one quantum-gravity measure, and a set of deliberately typed QFT/cosmology interfaces. *Sharper aliases* (current state of the reductions; the historical gate names are kept for ledger continuity): $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$ — the flavor ratios are *closed* (Plücker/Readout Rigidity, v49/v71/v75), what remains is *one dimensionful unit*, not a flavor gap; $(G_{\text{metric}}) \rightarrow G_{\text{net}}$ — reduced to a single boundary-net theorem, “*the seam–Calderón net is the index-4 μ_4 simple-current extension of the carrier net*” (holomorphy then follows, v89/GATE.METRIC.06); $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$ — named QFT/cosmology *transfer* interfaces (Koide source→pole, η_B Boltzmann, axion relic, m_p/m_e), never compiler powers. The status card:

Item	Status	Reading
parameter-freeness (self-consistency)	[I]/[L]	gapped transport $\Rightarrow$ <i>unique</i> Perron–Frobenius attractor (v56); bootstrap web $g_{\text{car}}=5$ forced (v6/v14)
Newton constant G_N	[I]/[P]	not <i>independent</i> : branch pins $\bar{M}_{\text{Pl}}$ rel. to Λ (v60); readout after branch + one dim. anchor
ratio $c_u/c_d = \frac{55}{117}$	[I]	$g_{\text{car}}\ \text{Pl}_{1,a}(K) \ _1 / (N_{\text{fam}}^2 \Delta_Q)$; rigid on the derived stratum (v49/v71), no solve
(U) flavor gate	[A] $\rightarrow v_{\text{geo}}$	Gate 1 closed: $U_{\text{point}} \rightarrow v_{\text{geo}}$ (ratios + Grand Mass Volume, v75); same anchor as $1/G$
$R + R^2$ gravity	[I]/[P]	covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)
physical QG sector	[F]/[P]	gap-decoupled admissible measure ($\Delta_{\text{eff}}=1.648>0$)
ambient QG measure	[A](reduced)	Gate 2 tier B: holographically reduced to a finite seam- <i>boundary</i> measure (v76); IR tier closed (decoupling); strict-TOE cert. open (G_{metric})
Koide	[P]	$Q_{\text{src}} + \varphi_0/24$ as source $\rightarrow$ pole transfer conjecture
axion DM	[P]/[A]	$f_a = M_{\text{scal}}/128$ as a scenario
η_B	[P]	readout closed; leptogenesis Boltzmann solve missing
m_p/m_e	[A]	cross-sector QCD/EW ratio
$N_\star$	[P]	reheating input, $N_\star \simeq 57$ (continuous)
$a = (1, 1, 2)$	[L]/[P]	forced by $4 = 2 + 1 + 1$, conditional on the carrier
E_8 Lean	[F] $\rightarrow$ [L]	hardening, not a blocker (script-certified)

Marker key: [I] *exact identity* · [L] *Lie/lattice theorem* · [F] *formalised* · [N] *numerical fixed point* · [P] *physical/conditional* · [A] *axiom / open*.

Only *two* research gates remain — (U_{wall}) and G_{metric} ; the frontier items are typed interfaces, not structural holes. Of these, (U_{wall}) is the only genuine *physics* gate left: G_{metric} is now heat-kernel grounded (G2: the spectral action gives $R + R^2$ structurally) and *gap-decoupled* (G5: $\Delta_{\text{eff}}=1.648>0$ isolates the closed admissible sector from the un-built ambient), so its residual — the ambient projective limit (G6) — is a *strict-TOE completion target*: its absence does not affect the bounded IR claim, but it does block certification as a strict physical TOE. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion `tfpt_research_contracts`, with priority (U_{wall}) first. `[verification/v36_spectral_action_g2.py]`

4 Before / after: what concretely changes

Aspect	State of Papers 1–7	After the compiler closure
E_8	<i>peripheral</i> : only an $11D \rightarrow 4D$ reduction aside and the nilpotent-orbit <i>cascade</i> (scale ordering)	central, constructed : $C^+ = D_5$ spinor, $\mathbb{P}^1 \setminus \mu_4 = A_3$, μ_4 glue $\Rightarrow E_8$ [L]
E_8 numbers	240, 248 read off tables when needed	$240 = 16 \cdot 5 \cdot 3$, $248 = 240 + 8$ <i>derived</i> as carrier traces [I]
fine structure α^{-1}	<i>already</i> a parameter-free cubic fixed point $\alpha^3 - 2c_3\alpha^2 - 8bc_6 \ln(\cdot) = 0$, $\alpha^{-1} \approx 137.0365$	<i>same</i> equation, now existence & uniqueness proved and refined to 137.0359992 ; dev. 2.9×10^{-10} vs CODATA-2022 (1.9σ) [I]/[N]
flavor residues	partly audit-flagged	anchor $\{1, 1, 2\}$ & matrix forced by 3 hypotheses [I]/[P]
many sectors	side by side, each its own closure	<i>one</i> compiler $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ generates them [I]
inputs	c_3, g_{car} (+ implicit structure)	exactly 2 axioms (P1, P2), clearly declared [A]
A_s, n_s, r (inflation)	scale present	scalaron mass fixed by c_3^7 (3.06×10^{13} GeV); A_s, n_s, r become <i>predictions</i> [P]
θ_{12} (solar)	open (leading $\frac{1}{3}$)	cond. derived $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.307$, given seam $\varepsilon = c_3$ [N]/[P]

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

Reduction in one line

The number of *structural* assumptions drops from “a fixed-point seed c_3 , a carrier, the orbit cascade, plus a per-sector calibration in each block” to **two axioms**. Everything else ($N_{\text{fam}}, \Omega_{\text{adm}}, b_1, \alpha^{-1}$, the flavor matrix, the E_8 glue, the scale grammar) becomes a *consequence*. *And even those two tighten*: the bootstrap note shows $g_{\text{car}} = 5$ and the integer 8 in c_3 are *overdetermined E_8 -closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5; $8 = h(D_5) = \text{rank } E_8 = \det R = \varphi(30)$), leaving *one* genuine continuous primitive: π from the Möbius boundary. So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.)

5 Paper by paper: what each original paper did, and how it works now

The seven original papers (plus the orientation note) total roughly **247 pages**. The table below puts each one next to its compiler-era replacement: *what* it treated and computed, *how many pages* it took, *where* that now lives and how much it compresses, and whether the *accuracy* of the result improves on top. The honest summary up front: in most sectors the *numbers are the same* — what shrinks dramatically is the *derivation length* and the number of independent assumptions; in three places (the solar angle, the inflation amplitude, the scalaron mass) there is a *genuinely new or sharper* result.

P	Treated & computed	Old	How it works now (note; compression)	Acc.
0	orientation: architecture, decoder core $Y, [u_\Sigma], u$, theorem stack	20 p	<i>one</i> compiler diagram: $\{c_3, g_{\text{car}}\} \rightarrow D_5 \oplus A_3 \rightarrow E_8$ (intro + action_grammar)	=
1	boundary kernel: $c_3 = \frac{1}{8\pi}$, Calderón polarization, carrier decoder $6Y^2 - Y - 1 = 0$	20 p	the two axioms P1 (c_3) + P2 ($g_{\text{car}}=5$); carrier = 3+2 atom (compiler)	=
2	carrier/SM packet: $S^+=16$, G_{phys} , $N_{\text{fam}}=3$, $\Omega_{\text{adm}}=48$, hypercharge, b_1	48 p	compiler reads $16=1+5+10$, N_{fam} , Ω_{adm} , $b_1 = \frac{41}{10}$ as carrier traces; $C^+ = D_5$ spinor (~ 5 p)	=
3	EM closure: α^{-1} via $\sum L + N_\Phi = 41$; flavor transport, CKM/PMNS, θ_{13}	38 p	α^{-1} as unique root of $F_{U(1)}(\alpha)=0$; flavor from transport theorem (parabolic + sm)	$\uparrow$
4	admissibility/strong CP/QFT: $\theta_{\text{eff}}=0$, OS reconstruction, mass gap	40 p	explicit gap $6 \log \frac{3}{2} \Rightarrow$ Dobrushin/OS; strong CP from structure+RP (closing_comp.)	=
5	geometric Hodge/metrolgy: M_{P1} , v_{geo} , Λ , spectral Einstein	30 p	Hodge/ R^2 branch; Λ <i>typed</i> (one branch label); boundary-normalised (closing_comp.)	=
6	cosmology interface: FRW, Λ_{IR} , scalaron, reheating, leptogenesis, axion	15 p	Starobinsky $M = c_3^{7/2} \bar{M}_{\text{P1}} = 3.06 \times 10^{13}$ GeV; A_s <i>parameter-free</i> (closing_comp.)	$\uparrow_{\text{new}}$
7	CMB operational closure: C_ℓ , $a_{\ell m}$, seam mixer, Planck pipeline	36 p	$n_s=0.964$, $r=0.0040$ from the R^2 attractor, no free amplitude (closing_comp.)	$\uparrow_{\text{new}}$

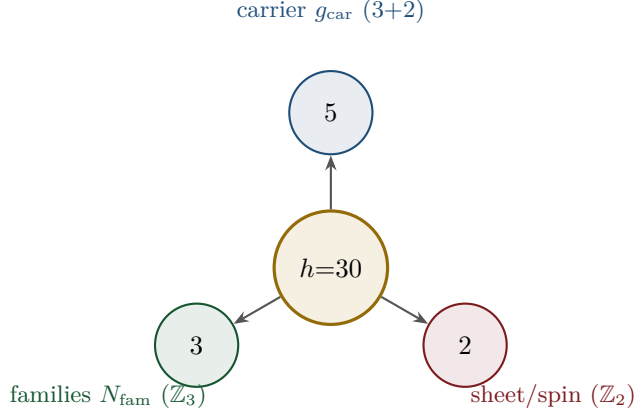
Three places where accuracy genuinely improves “on top”

1. **Solar angle θ_{12} .** Paper 3 left it as the only open SM number (leading $\frac{1}{3}$). It is now *conditionally derived* (modulo the seam-misalignment lemma): TBM gives $\frac{1}{3}$, the charged-lepton 1–2 misalignment is the seam $\varepsilon = \frac{N_{\text{fam}}}{|\mu_4|} \varphi_0^{\text{ret}} = \frac{3}{4} \varphi_0^{\text{ret}}$ (leading term exactly $c_3 = \frac{1}{8\pi}$), and TBM geometry gives $\sin^2 \theta_{12} = \frac{1}{3} - \frac{2}{3} \varepsilon = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.3067 - 0.1\%$ from NuFIT.
2. **Inflation amplitude A_s .** Generic Starobinsky fits the scalaron mass to A_s . TFPT fixes it by the seam, $(M/\bar{M}_{\text{P1}})^2 = c_3^7$, so $A_s = \frac{N_s^2}{24\pi^2} c_3^7 = 2.0 \times 10^{-9}$ is a *prediction* (Planck 2.1×10^{-9}), and the measured A_s even *predicts* $N_\star = 56$.
3. **Scalaron mass.** $M = c_3^{7/2} \bar{M}_{\text{P1}} = 3.06 \times 10^{13}$ GeV comes out *exactly* at the canonical Starobinsky value — a number that was an input before is now an output.

What “simpler” means concretely. The compression is not cosmetic. Paper 2 spent 48 pages establishing the carrier/SM packet by rigidity arguments; the compiler reads the same $16=1+5+10$, $N_{\text{fam}}=3$, $\Omega_{\text{adm}}=48$, $b_1 = \frac{41}{10}$ off the Pascal row of a 5-slot carrier in a page. Paper 3’s flavor transport (38 p) becomes one transport theorem plus a fixed residue matrix whose characteristic polynomial $t^3 - 9t^2 + 10t - 8$ carries only compiler numbers. The “many independent closures” of Papers 1–7 become *one* generator with a handful of corollaries.

6 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of E_8 is $h = 30 = 2 \cdot 3 \cdot 5$. Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:

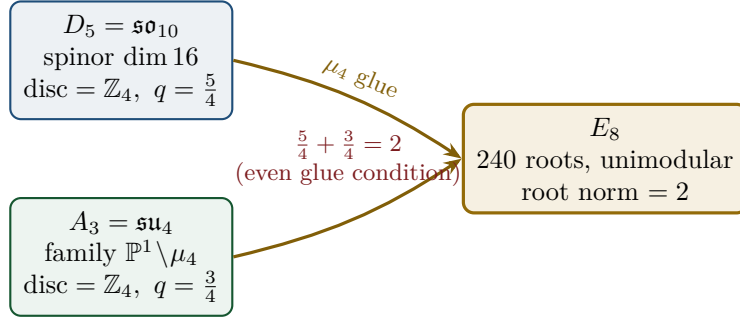


From this the compiler reads, among others:

- $\dim S^+ = 1 + 10 + 5 = 16$ (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ appears in family, action ladder and the E_8 root count. [I]
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$ and $\dim E_8 = 240 + (5 + 3) = 248$. [I]
- the $\mathbb{Z}_{30}$ Coxeter element $\leftrightarrow$ the cyclotomic polynomial Φ_{30} ; the corner rotation $z \mapsto iz$ on $\mathbb{P}^1 \setminus \mu_4$ is the A_3 Coxeter element. [I]

7 The μ_4 glue: how E_8 is really built

The decisive, now *proved* step: D_5 and A_3 have the *same* discriminant group $\mathbb{Z}_4$, and their discriminant-form norms are exactly two TFPT constants which add up to the E_8 root norm.



Glue theorem (proved)

$\text{disc}(D_5) = \text{disc}(A_3) = \mathbb{Z}_4$, glue index $|\mu_4| = 4$, and $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$ the E_8 root norm. Hence $E_8 = (D_5 \oplus A_3) + \mu_4\text{-glue}$ is a *closed lattice-theoretic* construction — not a blind “positing of 248”. [L]

8 How derivation now works: the derivation map

What can one concretely *compute* with $\{c_3, g_{\text{car}}\}$? The following map summarises what is now a consequence rather than an assumption.

Sector	How it now arises	Status
gauge group / families	carrier 3+2, dim $S^+=16$, $N_{\text{fam}}=3$, $\Omega_{\text{adm}}=48$	[I]
hypercharge	roots of the charge polynomial $bsY^2 + (s-b)Y - 1$	[I]
$U(1)$ index b_1	$b_1 = \frac{g_{\text{car}} 2^{g_{\text{car}}-2} + 1}{10} = \frac{41}{10}$	[I]
fine structure α^{-1}	unique root of $F_{U(1)}(\alpha) = 0$	[I]/[N]
flavor matrix	transport theorem (H1 proved, H2/H3 force the matrix)	[I]/[P]
electroweak scale	$v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$ (carrier divides α^{-1} by 5)	[I]
cosmological constant	$\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)	[I]/[P]
Hubble scale	$H_0 \sim \sqrt{\Lambda}$	[P]
inflation amplitude	$A_s = \frac{N_{\star}^2}{24\pi^2} c_3^7$ in the R^2 branch	[P]
E_8	$D_5 \oplus A_3 + \mu_4\text{-glue}$	[L]

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

Notable: the scale grammar is *one* exponential engine. The same number $\alpha^{-1} \approx 137$ generates the electroweak scale (divided by 5 = carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

Typing note on Λ (a deliberate danger zone). “ Λ ” denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent $e^{-2\alpha^{-1}}$ (an exact [I] readout), (ii) the dimensionless density quotient $\rho_{\Lambda}/\bar{M}_{\text{Pl}}^4$ (which carries a prefactor *branch*, hence [P]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

Cosmology is the Pascal action grammar of the carrier — not a separate module

With $x = v_{\text{geo}}/(5\beta_{\text{rad}}^2 \bar{M}_{\text{Pl}})$ the cosmological readouts are a *power tower* on the five-slot carrier graph K_5 :

$$\frac{H_0}{\bar{M}_{\text{Pl}}} = x^5 R_H, \quad \frac{\rho_{\Lambda}}{\bar{M}_{\text{Pl}}^4} = x^{10} R_{\Lambda}, \quad (1, 5, 10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}.$$

EW is the unit x^1 , Hubble is the product over the $\binom{5}{1} = 5$ vertices, Λ is the product over the $\binom{5}{2} = 10$ edges — the *same* Pascal triple that reads one generation, the action ladder and the E_8 root count $16 \cdot 5 \cdot 3 = 240$. “Hubble = slot product, Λ = pair product.” [I]

The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}} \lambda_Y^{L_{f,j}} \Lambda_{f,j}, \quad \lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1 - \varphi_0^{\text{ret}})}, \quad \varphi_0^{\text{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}.$$

The word-lengths $L_{f,j}$ are the *fixed residue matrix of the compiler*, whose characteristic polynomial $t^3 - 9t^2 + 10t - 8$ carries only compiler numbers ($\text{tr} = 9 = N_{\text{fam}}^2$, $\det = 8 = h(D_5)$, principal 2-minors 2,3,5 with product $30 = h(E_8)$). Through the φ_0^{ret} -ladder $\hat{m} = \frac{v_{\text{geo}}\pi}{\sqrt{2}} c(\varphi_0^{\text{ret}})^k$ all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, v_{49}/v_{71}), with only the *absolute* amplitude scale reducing to the (U_{wall}) anchor. **SM status:** the discrete

packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional; $m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$ are RG scheme-layer (m_H rests on the closed UV quartic $\frac{1}{16\pi^2}$); and the solar angle is *realized by a TFPT texture* and reduced to the seam-misalignment lemma ($\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$; see below). (*Full derivation: document 2, `tfpt_2_standard_model`.*)

9 Physical, geometric, topological: which solution is more plausible?

Level	E_8 peripheral (v4.5)	E_8 compiled (new)
physical	α^{-1} already a cubic fixed point, but each sector closed on its own	α^{-1} as the fixed point of <i>one</i> equation; one generator for many scales
geometric	Lie algebra external; E_8 only orders the cascade	the C^+ spinor + puncture geometry $\mathbb{P}^1 \setminus \mu_4$ give D_5, A_3 , glued to E_8
topological	—	the common discriminant $\mathbb{Z}_4$ (μ_4 winding) forces the glue; Φ_{30} /Coxeter drives the sectors
parsimony	many structures	2 axioms, the rest a consequence

Why the compiled solution is more plausible

It is (i) *more parsimonious* (fewer independent assumptions $\Rightarrow$ less overfitting risk), (ii) *more rigid* (the glue condition $5/4 + 3/4 = 2$ and the discriminant $\mathbb{Z}_4$ leave little freedom), and (iii) *more checkable* (machine-checkable E_8 appendix, explicit fixed-point equation for α^{-1}). More explained from less — exactly the criterion that makes a theory more probable.

10 What role E_8 now plays exactly

E_8 is no longer the starting point but the *output format*:

- **Bookkeeping:** E_8 is the unimodular container in which carrier (D_5) and family (A_3) fit together consistently. Its numbers 240, 248 are carrier traces, not inputs.
- **Consistency test:** unimodularity ($\det 1$) is a hard condition; it closes only because the discriminant-form norms $5/4$ and $3/4$ (both already present in TFPT) add up to 2. E_8 thus tests the internal coherence of the two atoms.
- **Not an end in itself:** E_8 does not explain “everything”; it is the smallest even unimodular hull of the datum $D_5 \oplus A_3$. The physics sits in the *two inputs*, not in E_8 .

New — the secondary atlas. Beyond that, E_8 is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (`tfpt_3_e8_audit_bootstrap`, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit:

$E_6 \times A_2$ reads the flavor matrix

$$248 = \|R\|_F^2 + \det R + 2(\mathbf{1}^\top R a) N_{\text{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2 = h(D_5)$. The discipline rule is: *every load-bearing number must appear in at least one E_8 projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

11 What this means for the theory-of-everything question

The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic $\leftrightarrow$ unitary representation); the algebraic selector R , the Q spectrum and the splitting type are fixed (now *derived*: $\det R = 8$ lattice, $\text{Spec}(Q_+)$ from D_4 , v69/v70), and the charged-lepton amplitudes are closed. The quark mass *ratios* ($c_u/c_d = \frac{55}{117}$ etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall (U_{point} , an anchor; Research Contract 1). [I]/[A]
2. **Gravity — exact in the R^2 regime.** Spectral action $\rightarrow R+R^2$, BRST/Hodge kills the gauge directions, FRW reduction $\rightarrow$ Starobinsky; the only remainder is the *controlled* low-curvature truncation ($R^{n \geq 3} \rightarrow R^2$, valid for curvature $\ll M^2$). [I] in the regime
3. **Strong CP — decoupled.** $\theta_{\text{eff}}=0$ follows from three structural facts (γ_5 -Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [I] (given RP)
4. **Full dynamics — the admissible transfer model is closed.** The mass gap is the *explicit* discrete gap $6 \log \frac{3}{2}$ of the C_6 transfer operator (eigenvalues $(k/3)^6$), which supports the OS reconstruction path on the admissible sector; the continuum and metric-coupled theory still require the stated RP, cluster and limit assumptions. [I] (transfer model) / [P] (continuum)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (1886 jobs, 0 sorry, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [I]/[A]

In other words: the theory has shrunk from “many open audit points” to “*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms”. The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. (*Details and proof chains: document 2, `tfpt_2_standard_model`, Part III.*)

The TOE-completion ledger — what is done, and what genuinely remains

A theory-of-everything must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

TOE piece	Status	If not complete: what closes it
gauge group, N_{fam} , hypercharge, b_1	[I]	— complete
$E_8 = (D_5 \oplus A_3) + \mu_4$, group closure	[L]	— complete
bootstrap $(g, \mu) = (5, 4)$, “8” in c_3	[L]	— complete (reverse glue $\mu^2 - 5\mu + 4 = 0$)
fine structure α^{-1}	[N]	— existence+uniqueness proved ; value is a numerical fixed point (ledger EM.FP.01), Ward origin [P]
fermion masses (φ_0^{ret} -ladder)	[I]/[A]	leptons exact; quark <i>ratios</i> closed (Plücker, v49/v71); absolute scale = anchor
mixings $\theta_{13}, \theta_{23}, \theta_{12}$	[N]/[P]	θ_{12} cond. derived; θ_{23} maximal
flavour matrix / H2	[I]	up to Mehta–Seshadri unitarity (U), a Paper-3 input
inflation A_s, n_s, r , scalaron M	[N]	parameter-free given $M_{\text{Star}} = M_{\text{scal}}$
<i>genuinely open — the actual TOE remainder</i>		
dim. EW/QCD masses	[P]	standard RG threshold matching (no new physics)
QFT measure (admissible sector)	[I]	closed: RP from cusp spectrum, OS reconstruction
ambient QFT + full gravity (metric sector)	[A]	<i>not constructed here</i> (frontier doc); conditionally reduces to one named hypothesis (ambient RP) on the relative spectral-action measure
analytic boundary origin of π ($c_3 = \frac{1}{8\pi}$)	[P]	hardened: $c_3 = 1/(\mathbb{Z}_2 \oint_{S^2} K dA) = 1/(8\pi)$ (Gauss–Bonnet); “8” = rank E_8 ; π stays the one primitive
$\eta_B, m_p/m_e$, Koide, dark matter	[A]	frontier handles only (see document 4)

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

The honest bottom line: the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one φ_0^{ret} -ladder (lepton rationals exact, quark $O(1)$ digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed); the *ambient/metric-sector* measure — i.e. *full quantum gravity* — is *not* constructed here and stays [A] (frontier doc), reducing conditionally to a single named hypothesis (ambient reflection positivity). The seam seed is *hardened*: $c_3 = 1/(|\mathbb{Z}_2| \oint_{S^2} K dA) = 1/(8\pi)$ is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete “8” equal to rank E_8 and only the Gauss–Bonnet π left as the single irreducible primitive. What remains for a full TOE is then the ambient-RP hypothesis (full QG), plus the standard-RG dimensionful masses and the honest frontier items. No load-bearing discrete parameter remains free, and π is the lone continuous primitive.

12 Does this make the theory more probable?

Plausibility balance

Pro: (a) two inputs instead of many; (b) α^{-1} to 2.9×10^{-10} (CODATA-2022, 1.9σ) as a *fixed point*, not a fit; (c) E_8 constructed, not postulated; (d) one compiler generates several sectors $\Rightarrow$ fewer degrees of freedom, more predictive power.

Contra/open: what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ($m_W, m_Z, m_H, \sin^2 \theta_W, \alpha_s$) sit on the RG

scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ($\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$, modulo the seam-misalignment lemma). A final verdict “it maps reality” requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

13 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

Observable	TFPT value & derivation	Experiment (running / upcoming)	St.
solar angle $\sin^2 \theta_{12}$	$\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.3067$; TBM + seam misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\text{ret}} \approx c_3$ (cond.)	JUNO (taking data since Aug 2025; sub-percent θ_{12}) — the sharpest live test	[N]/[P]
reactor angle $\sin^2 \theta_{13}$	$\varphi_0^{\text{ret}} e^{-5/6} = 0.0231$; seed $\times$ carrier-trace $e^{-\gamma}$, $\gamma = \frac{5}{6}$	Daya Bay / RENO / JUNO (PDG 0.0220)	[I]
atmospheric $\sin^2 \theta_{23}$	$\approx \frac{1}{2}$ ($\mu\tau$ -symmetric limit; octant not selected)	NOvA / T2K / DUNE (octant ambiguity, NuFIT 6.0)	[P]
fine structure α^{-1}	137.0359992 as the unique root of $F_{U(1)}(\alpha)=0$ (no fit, no second branch)	CODATA-2022 137.035999177(21): dev. 2.9×10^{-10} from the recommended adjustment (1.9 σ of its uncertainty)	[I]/[N]
tensor ratio r	$r = \frac{12}{N_*^2} = 0.0040$ ($N_*=55$); R^2 attractor, $M=c_3^{7/2}\bar{M}_{\text{Pl}}$	now $r < 0.036$ (BICEP/Keck BK18); CMB-S4 ($\sigma_r \leq 5 \times 10^{-4}$, obs. ~ 2033)	[P]
tilt n_s	$n_s = 1 - \frac{2}{N_*} = 0.964$ (same attractor)	Planck (0.965); CMB-S4 sharpens	[P]
amplitude A_s	$\frac{N_*^2}{24\pi^2} c_3^7 = 2.0 \times 10^{-9}$, parameter-free	Planck (2.1×10^{-9})	[P]
neutron EDM	$\theta_{\text{eff}} = 0 \Rightarrow d_n$ essentially null (structure + RP)	PSI nEDM / SNS (running)	[I]
$0\nu\beta\beta$ / ordering	Majorana branch; normal-ordering preference	LEGEND / nEXO , DUNE, KATRIN	[P]
flavor invariants	$\det R = h(D_5)=8$, principal minors (2, 3, 5), $\chi_R = t^3 - 9t^2 + 10t - 8$ (exact)	any future CKM/PMNS global fit must satisfy these	[I]
H_0 vs. Λ	$v_{\text{EW}} \sim e^{-\alpha^{-1}/5}$, $\Lambda \sim e^{-2\alpha^{-1}}$, $H_0 \sim \sqrt{\Lambda}$: one engine	SH0ES / DESI / Planck (Hubble tension)	[P]

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

The two decisive near-term tests

- (1) **JUNO and θ_{12} .** JUNO is *taking data since August 2025* and targets sub-percent $\sin^2 \theta_{12}$. TFPT commits to the conditional $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.3067$ (seam $\varepsilon = c_3$). A measured central value away from ~ 0.307 at high significance falsifies the seam mechanism — a *live* test.
- (2) **r .** The R^2 scalaron with M fixed by c_3^7 predicts $r = 0.0040$ ($N_*=55$), already below the current bound $r < 0.036$ (BICEP/Keck BK18) and within reach of CMB-S4 ($\sigma_r \leq 5 \times 10^{-4}$, first observations ~ 2033 –34). A detection of $r \gtrsim 0.01$ would be incompatible with the R^2 branch on which M_{Pl} and A_s rest.

Structural (non-experimental) checks. Independent of any apparatus, the machine-checkable E_8 appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two

TFPT atoms really generate E_8 ; and the discipline rule “*every load-bearing number appears in at least one E_8 projection*” (atlas note) turns the number stock into a falsifiable raster rather than numerology.

Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar** $\sin^2 \theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$ — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor** $r \approx 0.0040$ — any robust $r \gtrsim 0.01$ kills the R^2 branch carrying M_{Pl}, A_s ;
- **ordering** normal, small $m_{\beta\beta}$ — inverted ordering or large $m_{\beta\beta}$ kills the Majorana branch;
- **strong CP** $\theta_{\text{eff}} = 0$ — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$ — a robust $w \neq -1$ kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

Blind-prediction registry (frozen 2026-06-09, machine-enforced). Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [`verification/v84_frozen_registry.py`] re-derives each decimal from the two axioms on every suite run (formula $\leftrightarrow$ value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one* θ_{12} prediction of record (the seed $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards. r and n_s enter only as *bands* over the declared external $N_\star \in [50, 60]$, never as point predictions.

What likely follows next

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, v49/v69/v71); the only residual is the *absolute* amplitude scale, an anchor (U_{point}) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to $b_1 = 41/10$), since the compiler systematically produces traces over S^+ ;
- a sharper geometry $\leftrightarrow$ gravity bridge once the Hodge/ R^2 branch is lifted from [P] to [I].

Conclusion

The state of Papers 1–7 was an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ builds, from two inputs $\{c_3, g_{\text{car}}\}$ via $D_5 \oplus A_3 + \mu_4$, the E_8 container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable*. Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

Appendix: computational verification (reproducibility)

Every claim in this document marked [\[I\]](#) (exact identity), [\[L\]](#) (Lie/lattice theorem) or [\[N\]](#) (numerical fixed point) is re-derived from the two axioms $\{c_3, g_{\text{car}}\}$ alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references `[verification/...]` throughout the documents point to the exact script; the table below is the master map.

Script	What it machine-checks
v1_e8_glue.py	glue theorem $E_8=(D_5\oplus A_3)+\mu_4$: disc $=\mathbb{Z}_4$, $q(D_5)+q(A_3)=\frac{5}{4}+\frac{3}{4}=2$, $240=16\cdot 5\cdot 3$, $248=240+8$
v2_carrier_pascal.py	$g_{\text{car}}=5$ unique Pascal solution; $16=1+5+10$, $N_{\text{fam}}=3$, rank $E_8=8$, $\Omega_{\text{adm}}=48$, $b_1=\frac{41}{10}$
v3_em_alpha.py	EM closure: unique root of $F_{U(1)}(\alpha)=0$, $\alpha^{-1}=137.0359992168$, existence/uniqueness, inverse test, ablation
v4_flavor_matrix.py	residue matrix R : det $=8$, minors $\{2, 3, 5\}$, $\chi_R=t^3-9t^2+10t-8$, SNF diag $(1, 1, 8)$, $\ R\ _F^2=78$, $\sum L=40$
v5_e8_cascade.py	cascade $D_n=60-2n$: endpoints, exponent rungs $\rightarrow 240$, IR tail 46080, variance 6552
v6_bootstrap.py	reverse glue $\mu^2-5\mu+4=0$; $g_{\text{car}}=5$ three ways; $8=\text{rank } E_8=h(D_5)=\varphi(30)$
v7_gravity_cosmo.py	scalaron exponent 7, $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2=c_3^7$, A_s, n_s, r , Ω_b , $\sin^2 \theta_{12}$
v8_horizon.py	horizon code: $\frac{1}{2\pi}=4c_3$, $1920= W(D_5) $, S_{dS} , Page, $\beta_{\text{rad}}=0.2424^\circ$
v9_neutrino_texture.py	polar angle: explicit $\mu\tau$ Majorana texture $\rightarrow \sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$, $\theta_{23}=45^\circ$, $\theta_{13}=0$
v10_projection_involutions.py	Q, Σ algebra: $K=R+Q\Sigma$, $L=R+Q(I+\Sigma)$; χ_Q, χ_K ; det ladder $3\cdot 4\cdot 8\cdot 20=1920= W(D_5) $; $a^\top K a=41$
v11_unique_KQ.py	K, Q are the <i>unique</i> nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)
v12_mass_generation_polynomials.py	sector/generation polynomials of K and the anchor-block det ladder $(9, 10, 16, 40)$
v13_open_gates.py	gate closures: $M=41=10b_1=\sum L+N_\Phi$; $Q_+=A_3$ exponents, $Q_-^2=N_{\text{fam}}$
v14_carrier_uniqueness.py	$g_{\text{car}}=5$ unique ($N_{\text{fam}}\in\mathbb{Z}^+$, rank $E_8=8$); split $(3, 2)$ from $b+s=5$, $b^2+s^2=13$; $\text{Tr } Y=\text{Tr } Y^3=0$
v15_bootstrap_classification.py	$D_5\oplus A_3$ unique familyful cyclic-gluon of E_8 (16-spinor + 3 families), glue $\mathbb{Z}_4$
v16_solar_dual_anchor.py	$a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2}, -\frac{1}{2}, 1)$; $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$; full PMNS
v17_hexagonal_resolvent.py	finite resolvent $D_y^{-1}=(\sum y^{5-m}\delta^m U_6^m)/(y^6-\delta^6)$ (quark c backbone)
v18_quark_yukawa.py	quark source ratios $\frac{m_u}{m_d}=0.470085$, $\frac{m_c}{m_s}=13.61$, $\frac{m_t}{m_b}=40.80$ exact; lepton $c=(\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$; full hierarchy
v19_monodromy_moduli.py	exact pole $z_\star=\frac{794-7\sqrt{9961}}{2187}$ (3^7); D_4 $SU(3)$ monodromy constructible but D_4 -fixed locus positive-dim $\Rightarrow$ exact quark c reduce to (U)
v20_lepton_c_derivation.py	lepton c 's derived: $\delta=\frac{1}{2}$ resolvent $c= \mu_4 ^w/(\frac{5}{4}-\cos\frac{r\pi}{3})\Rightarrow\frac{16}{7}, \frac{4}{3}$; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}=\frac{32}{9}\Rightarrow\frac{7}{6}$
v21_solar_product_quark.py	solar coeff $=q(A_3)=\frac{3}{4}$ (glue-norm, $q(D_5)+q(A_3)=2$), $\sin^2 \theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\text{ret}}}{2}$; product $\frac{2^{g_{\text{car}}}}{N_{\text{fam}}^2}$; quark law fails ($\frac{1}{7}$ vs 0.724)
v22_open_gates_audit.py	residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)
v23_anchor_generator.py	anchor $a=(1, 1, 2)$: $e_k=(4, 5, 2)$, $c_3=\frac{1}{2e_1\pi}$; $p_n=2+2^n\Rightarrow 240, 248$, binary ladder; inputs $\rightarrow \{a, \pi\}$
v24_quark_ratio_closure.py	quark ratios $\frac{55}{117}, \frac{34}{47}, \frac{3}{26}$ from TFPT blocks; mass ratios match $< 0.03\%$; rational c gauge, Λ_q in $O(1)$
v25_frontier_conjectures.py	conjectures: Koide $Q_{\text{src}}+\frac{\varphi_0^{\text{ret}}}{24}\approx\frac{2}{3}$; axion $f_a=M_{\text{scal}}/128\rightarrow m_a\approx 23.8\mu\text{eV}$
v26_flavor_frontier_unification.py	the “11” is not uniquely forced ($\Rightarrow$ [P]); flavor frontier unifies to one (U) gate; cusp config on the parabolic stability wall
v27_wall_representative.py	explicit balanced wall rep W_{wall} (cols = perm $(0, 1, 2)$, rows $w=(2, 1, 1)$, pardeg 0); selector det $R=8$, Spec $(Q_+)=\{1, 2, 3\}$
v28_gravity_fR.py	$R+R^2$ closed: $f(R)$, $M=c_3^{7/2}\bar{M}_{\text{Pl}}$ scalaron, $N_\star=57\rightarrow n_s, r, A_s$; full metric measure open; $248c_3^2<6\log\frac{3}{2}$
v29_research_contract_	research-contract certificates: $C_U^{(1)}$ wall enumeration $(1296\rightarrow 144\rightarrow 5)$ orbits; symmetry alone does not pick W_{wall} and $G5$ gap-dominance $2\ V\ <\Delta$

How to run. From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim $\leftrightarrow$ script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all six documents are written against). The [F] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).